Hit or Miss

Understanding Emergence and Absence of Homo-oligomeric Contacts in Protein Language Models

Zhidian Zhang et al. — bioRxiv, 2025

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Can one sequence reveal contacts between identical chains?

A homo-oligomer contains multiple copies of the **same protein sequence**.

- Selection preserves both the monomer fold and contacts between subunits.
- A single-sequence protein language model (pLM) may therefore encode both types of constraint.
- A predicted contact that looks wrong within one chain may be correct across two copies.

The paper's question
When do existing pLMs recover these interface contacts, and what explains their absence?

The contribution is an analysis of existing models and their failures, rather than a new assembly predictor.

Introduction; Results 2.1.

A contact map collapses chain identity

Let i, j be positions in one sequence.

- Monomeric: i and j touch within chain A.
- Interface: i in A touches j in B.
- Shared: both relationships hold.

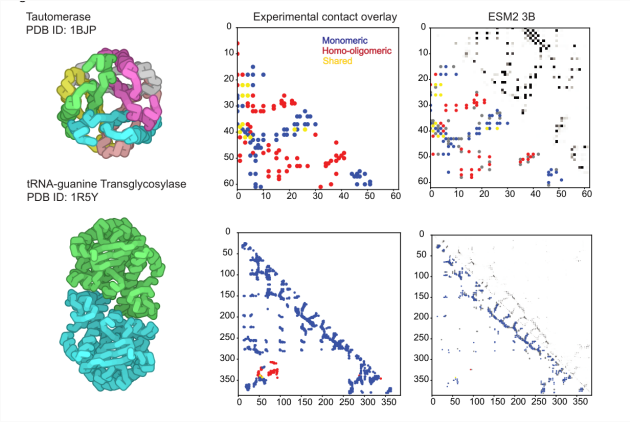
A model outputs an $L \times L$ map, without specifying which physical chain supplies each residue.

Illustrative pair: positions 12 and 47
Far apart inside A, but adjacent across A–B. A high model score can reflect assembly constraints rather than a folding error.

This is a toy example, not a measured residue pair from the paper.

Concept adapted from Fig. 1B; Methods 4.5.

An existence proof and a counterexample



1BJP, tautomerase: ESM2-3B recovers contacts explained by the homo-hexamer. **1R5Y,**

tRNA-guanine transglycosylase: interface signal is weak despite an oligomeric structure. Blue:

monomeric; red: interface; yellow: shared. Black points show model predictions.

Zhang et al., Fig. 1C (cropped). Input to ESM2 is one monomer sequence.

Three datasets answer different questions

Dataset	Construction	Purpose
4,160 PDB assemblies	30% sequence clustering, then Foldseek; assembly 1	How common are missing interface signals?
EPPIC: 40 biological, 75 crystal interfaces	Size-challenging curated sets; ambiguous or contact-free cases excluded	Scaling, contact-head refitting, biological versus crystal contacts
537 ProtCID-derived assemblies	Families with multiple interface clusters	Does restricting evolutionary context restore interface signals?

PDB curation excludes ≥ 10 identical chains, resolution > 5 Å, and chains outside 20–500 resolved residues. This is not an unrestricted proteome benchmark.

Methods 4.1, 4.4, 4.7. ProtCID screening starts from 411 families / 736 processed structures.

How recovery is measured: P@X

ConFind defines contacts by contact degree > 0.01; only pairs with sequence separation ≥ 6 are scored.

$$\mathrm{P@X}_{\mathrm{type}} = \frac{\#\text{ true contacts of that type among top } X \text{ scores}}{\#\text{ true contacts of that type}}$$

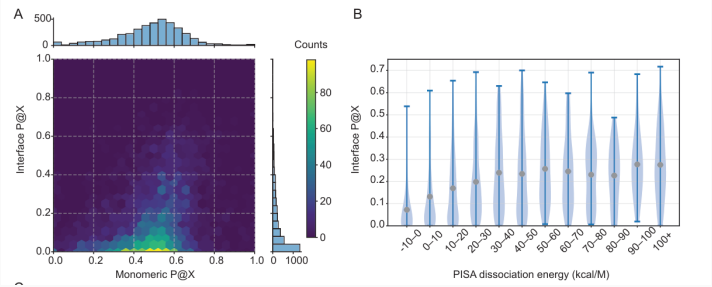
The evaluation separates **interface-exclusive** and **monomer-exclusive** contacts to avoid counting overlap as interface-specific evidence.

Toy calculation
If there are 20 interface-exclusive contacts, rank predictions at the prescribed true-contact budget. Recovering 8 of those 20 gives interface P@X $= 0.40$.

This uses a ground-truth-informed rank budget. It does not provide a deployment threshold, calibrated probability, or full precision–recall curve.

Methods 4.5. Text describes X using the ground-truth contact count; Fig. 1 uses the contact overlay budget.

Interface recovery is lower and highly variable



Good monomeric recovery does not ensure good interface recovery. Very low

PISA dissociation energy is associated with reduced interface recovery. Large

variation remains at other energies: stability alone is insufficient.

Fig. 2A–B (cropped); Results 2.2. PISA analysis available for 1,291 selected assemblies.

Some apparent misses reflect uncertain assembly labels

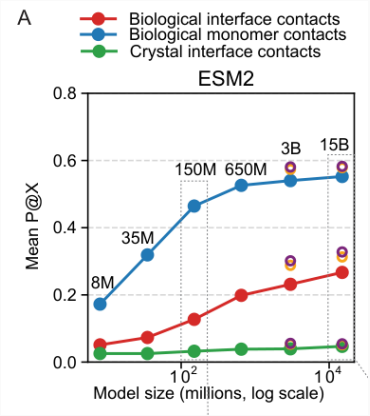
The authors linked structures to papers, expanded protein aliases, and used a GPT-4o evidence-extraction pipeline to identify solution-state measurements.

ESM2-15B interface signal	Papers with evidence	Reported monomer
No recovered interface contact	55	20 (36.4%)
High recovery: P@X ≥ 0.4	60	6 (10.0%)

Example: YfiH, 1XAF. Deposited as a dimer; its primary paper reports a monomer by size-exclusion chromatography. Interpretation: missing signal can flag a label conflict. The selected, accessible-paper sample cannot estimate the error rate of the whole PDB.

Fig. 2C–E; Methods 4.2–4.3. Initial groups: 147 zero-signal and 196 high-recovery entries.

Scaling strengthens interfaces after folds begin to plateau



On the curated EPPIC sets:

- Increasing ESM2 size improves interface recovery.
- Monomeric gains become smaller at large model sizes.
- Crystal-contact recovery remains near zero.

Refitted heads are shown as open circles. The backbone learns from sequences, but the **contact readout is structure-supervised**.

Fig. 3A (cropped); Results 2.3. Original ESM2 contact head trained using 20 PDB structures.

Scaling example: the fold is learned before the interface

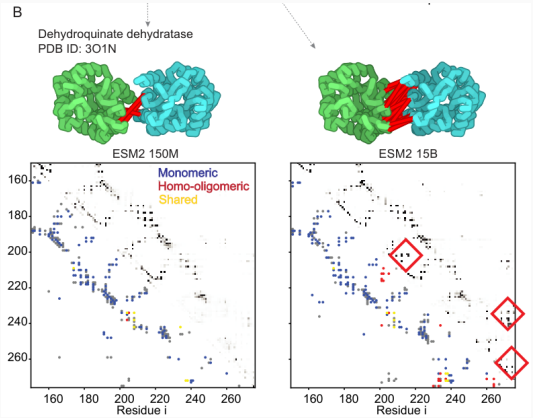


Fig. 3B (cropped); Results 2.3.

3O1N: dehydroquinate dehydratase. ESM2-150M

captures many monomer contacts but little interface signal. At 15B,

extra predictions align with the biological interface (red diamonds). This illustrates the aggregate trend;

it does not establish a universal parameter threshold for emergence.

A better readout recovers useful interface constraints

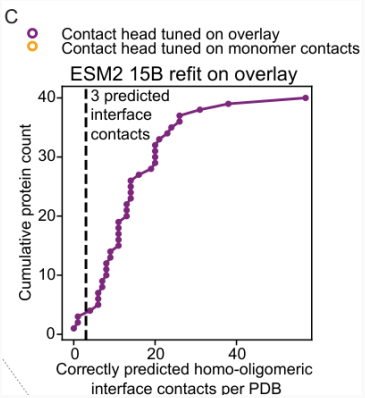


Fig. 3C (cropped); Methods 4.6; Results 2.3.

Refit the 3B/15B contact head on ConFind contacts, using either monomer labels or their overlay with interface labels. Exclude training assemblies in the same 30% PDB sequence cluster as the EPPIC test set. With the overlay-refit 15B head, **37/40** biological interfaces have ≥ 3 correctly recovered interface pairs. Limit: these are correct pairs identified retrospectively. The paper does not validate an end-to-end docking pipeline or show how to select only correct constraints prospectively.

Biological versus crystal contacts: a promising discriminator

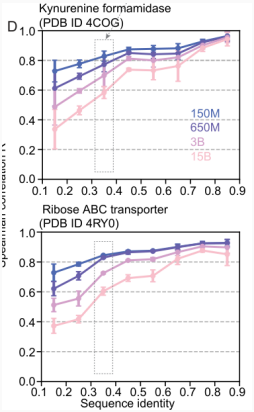
EPPIC deliberately includes large crystal contacts (> 1000 Å^2) and smaller biological interfaces (< 2000 Å^2), making interface size an incomplete shortcut.

- Biological interfaces have stronger recovered sequence constraints than crystal interfaces (Fig. 3A).
- Three crystal-set exceptions are 1WLY, 1WOQ and 2OW9.
- The current EPPIC server calls their assembly 1 biological; PISA also assigns oligomers for 1WLY and 2OW9.

Evidence level
Agreement with other computational annotations supports reinspection. It does not replace independent solution experiments or constitute a validated clinical-style classifier with a fixed threshold.

Results 2.3; Methods 4.4. EPPIC evaluation: 40 biological and 75 crystal interfaces.

A mechanistic hypothesis: larger models separate subfamilies



The categorical Jacobian records how changing one residue changes model outputs at other positions: $J \in \mathbb{R}^{L\times 20\times L\times 20}$. For aligned query–homolog pairs, retain top-L coupled residue pairs and strong Jacobian entries; compare them by Spearman correlation. At 30–40% sequence identity, correlations are roughly 0.8 for 150M/650M and 0.6 for 15B. **Authors' hypothesis:** finer model distinctions separate interface-specific subfamilies. Lower correlation alone does not prove explicit internal clustering.

Fig. 3D (two example families); Methods 4.8; Results 2.4.

Control the homolog context with MSA Pairformer

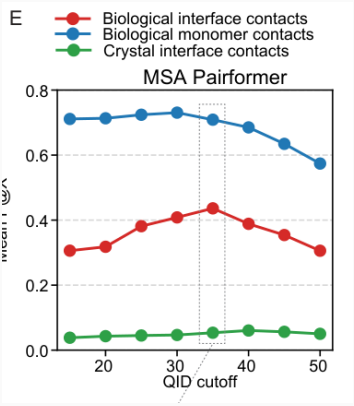
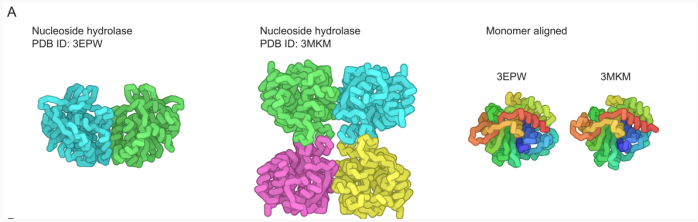


Fig. 3E (cropped); Results 2.4; Methods 4.7.

MSA Pairformer: 111M parameters; query-biased attention disabled so the MSA directly controls context. HHblits alignments, at most 512 sequences.

Raise the minimum identity to the query (QID) to remove distant homologs. At broad QID 15%, interface recovery is comparable to ESM2-15B. At QID 35%, mean interface P@X increases **from 0.33 to 0.44**. The result concerns interface recovery, not complete structure accuracy or a controlled compute comparison.

Why would fewer homologs help? A running example



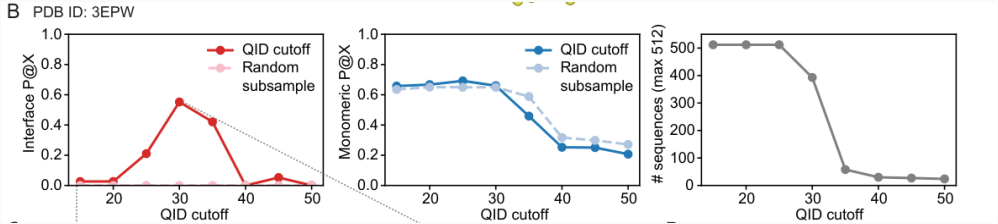
3EPW and 3MKM: nucleoside hydrolases. Their monomers share a fold, while their assemblies use different interfaces.

A broad alignment mixes constraints supporting different assemblies. Close homologs may preserve the query's interface more consistently.

Toy analogy: two subfamilies preserve the same core contacts but couple different surface residue pairs. Pooling them retains the fold signal while diluting either interface signal.

Fig. 4A (cropped); Results 2.4. Toy analogy is explanatory, not a fitted generative model.

3EPW: identity filtering outperforms equal-depth sampling

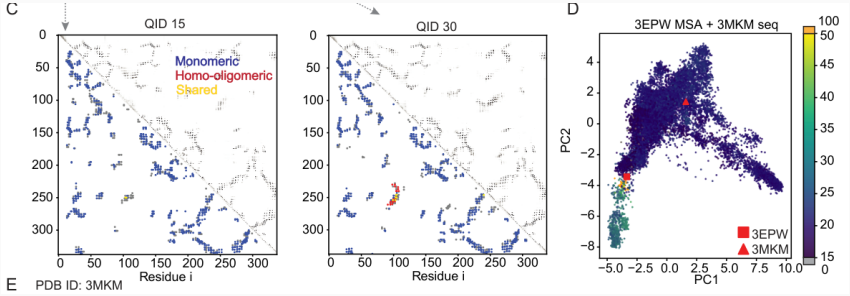


Interface recovery peaks near QID 30%; matched-count random subsampling does not reproduce the gain.

Monomer recovery is initially stable but degrades when too few sequences remain. The optimal context is a balance between subfamily consistency and evolutionary information.

Fig. 4B (cropped); Results 2.4. Red: interface; blue: monomeric. Lighter curves: random control.

3EPW: recovered pairs and sequence context

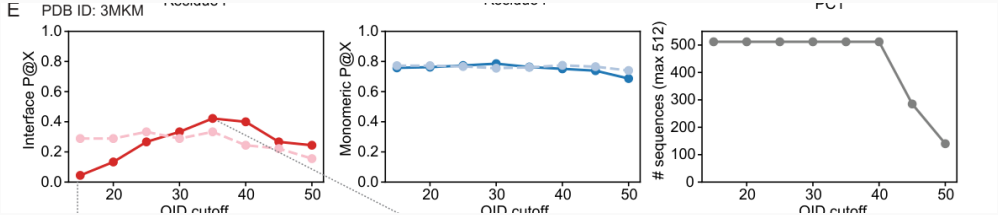


At QID 30%, extra predicted pairs match the known interface. PCA places the two reference proteins in different regions of the alignment's sequence space.

The PCA plot is descriptive: it does not establish a causal interface cluster boundary.

Fig. 4C–D (cropped). Left/middle: contact maps; right: MSA PCA colored by identity to query.

3MKM: selective context helps until information runs out

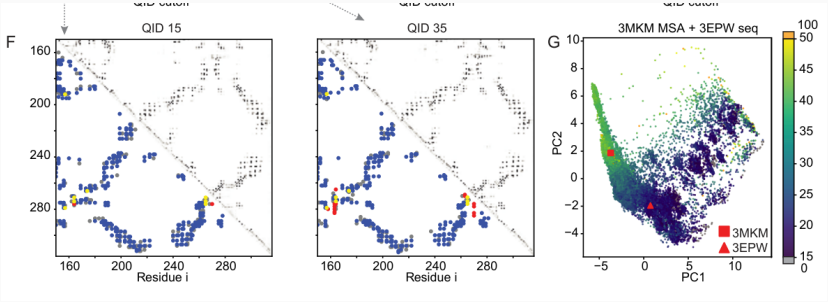


Interface recovery improves up to about QID 35%, while monomer recovery changes little over much of the scan.

At stricter cutoffs, depth collapses and interface recovery declines. Closer homologs are not always better if the alignment loses too much information.

Fig. 4E (cropped); Results 2.4. Matched-count random sampling controls for alignment depth.

3MKM: a distinct interface with a conserved monomer fold

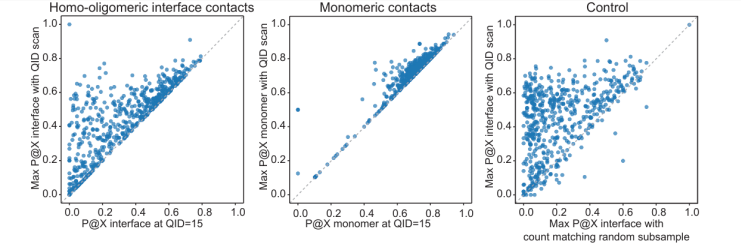


The QID 35% map recovers more interface contacts than the QID 15% map. The PCA plot again separates the reference proteins in sequence space.

Together, the two examples support dilution of interface-specific constraints when an alignment mixes divergent assemblies.

Fig. 4F–G (cropped); Results 2.4.

Generalization across families with multiple interfaces



Across 537 evaluated assemblies, the best QID scan frequently improves interface recovery with smaller changes in monomer recovery. Equal-depth random sampling is a relevant control.

Selection caveat: "best QID" is selected using known structural contacts. This is an upper-bound analysis of potential benefit, not a deployable rule for choosing QID for an unknown structure.

Fig. 5 (cropped); Methods 4.7. Families selected for multiple ProtCID interface clusters.

What a missing signal can and cannot tell us

Possible explanation	Evidence or next check
Assembly label conflict	Inspect solution-state experiments and conditions; YfiH illustrates this route.
Interface heterogeneity	Compare related structures and refine MSA context; 3EPW/3MKM provide an intervention.
Weak evolutionary constraint	Check transient or condition-dependent oligomerization; PISA association is suggestive.
Insufficient model/data	Compare scale, depth and readout; a true interface may remain invisible.

Absence is ambiguous. Even perfect fold recovery cannot establish that an interface is biologically absent.

Synthesis of Results 2.2–2.4 and Discussion; next checks are journal-club interpretation.

Discussion: turning an evolutionary signal into a tool

1. **A useful result:** single-sequence models encode interface constraints; scaling and controlled homolog selection can expose them.
2. **An unresolved mechanism:** does model scaling specifically separate oligomeric subfamilies, or simply improve conditional sequence modeling more broadly?
3. **A practical next experiment:** choose QID without structural labels, then evaluate on held-out families with solution-validated assemblies and actual docking outcomes.

Take-home message
A pLM "false contact" can describe quaternary structure, and a missing interface signal can be biologically informative. Both require evidence beyond the contact map.

Zhang et al., bioRxiv 2025.11.16.688745v1. Full paper + Methods read; no separate supplement linked.